

MENTIS · Procurement Intelligence Brief

Substance: 0.9% saline

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Executive Summary

0.9% saline, commonly known as normal saline, is an isotonic solution primarily used for hydration, electrolyte balance, and as a carrier for intravenous drug delivery in clinical settings. The market for this solution is valued at \$1.2 billion globally as of 2024, with a projected CAGR of 5.5% through 2034 [source](#). Regulatory compliance, including stringent sterility standards and FDA oversight, influences market stability [source](#). Key supplier facts highlight dominance by Baxter International, leveraging advanced packaging, and the Kelun Group, driving Asian market growth [source](#). Procurement strategies should consider price discrepancies, bulk purchasing, and emerging risks like kidney injury associated with prolonged use [source](#).

Product Profile

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Clinical Use & Mechanism

The clinical use of 0.9% saline spans a wide variety of therapeutic applications, particularly in perioperative and emergency settings. For instance, it plays a critical role in postoperative recovery protocols, particularly for procedures such as caesarean delivery. Recent research indicates that multimodal analgesia, which includes the use of 0.9% saline for wound infiltration with local anesthetics, can enhance recovery while minimizing systemic opioid exposure [source](#). However, robust evidence on its effectiveness in unplanned caesarean sections remains an area for further inquiry.

Additionally, 0.9% saline solutions are often utilized to alleviate postoperative sore throat (POST), a common complication linked to endotracheal intubation during surgeries requiring double-lumen tubes. Studies suggest that coupling saline-based delivery with local anesthetic techniques can aid in managing both hemodynamic instability and discomfort in such scenarios [source](#).

Another critical clinical application of saline lies in emergency settings, particularly for managing ureterolithiasis, which causes severe flank pain in patients presenting to the Emergency Department (ED). Given the drawbacks of opioid-based pain management, saline solutions are part of ongoing explorations into alternative therapeutics for this condition [source](#).

Conclusively, while 0.9% saline serves as a cornerstone in various clinical protocols, gaps in research—such as its specific efficacy in unplanned surgical cases—highlight opportunities for further studies to optimize its application across medical disciplines.

Market Size & Demand Drivers

The market demand for 0.9% sodium chloride sterile saline is robust and driven by diverse applications in healthcare. Globally, this market was valued at approximately \$1.2 billion in 2024, with projections suggesting a steady growth to \$1.9 billion by 2034, representing a Compound Annual Growth Rate (CAGR) of 5.5% over the forecast period [source](#). This growth is underpinned by its entrenched use across hospitals worldwide, where isotonic 0.9% saline dominates with a 71.7% market share due to its established position in medical protocols for hydration, electrolyte balance, and intravenous drug delivery [source](#).

Prefilled, single-dose sterile saline solutions held a dominant 68% market share by 2025, reflecting growing demand for safer, rapid administration in emergency and outpatient settings. Technological advancements in sterilization and barrier materials have further bolstered preference for this product type over multi-dose vials, which are becoming less prominent [source](#). Globally, over 1 million liters of 0.9% saline are administered daily, underscoring its critical role in clinical care [source](#).

However, competition from balanced and alternative formulations poses challenges. While traditional isotonic saline maintains leadership, balanced solutions are growing rapidly at a 9.4% CAGR, driven by their positioning as superior options for conditions like sepsis and diabetic ketoacidosis [source](#). This dynamic suggests the 0.9% saline market will need to innovate to retain its prominence in a competitive landscape.

Top Manufacturers & Suppliers

The market for 0.9% saline solution, also referred to as normal saline, is dominated by several key manufacturers globally. Baxter International leads the sector, leveraging its extensive manufacturing network and innovative packaging solutions such as the VIAFLEX flexible container system, which has gained significant traction in hospital settings [Baxter International Market Leadership](#). Baxter has additionally expanded its manufacturing footprint, receiving FDA approval for its Sabinanigo, Spain facility to supply the U.S. market [Baxter Facility Approval](#).

In Asia, the Kelun Group from China emerges as a dominant player, supported by its extensive domestic footprint and exports to over 80 countries [China's Kelun Group](#). Riding on substantial cost advantages and growing healthcare infrastructure, the region's saline production is experiencing robust expansion, with India and China posting CAGRs of 8.9% and 8.6% respectively [Asia-Pacific Growth](#).

In addition to established players, UFC Bio caters to niche applications like laboratory research and nasal irrigation, ensuring quality raw materials for contamination control in scientific environments [UFC Bio Applications](#).

The global market demonstrates diversification by region—with North America holding a 46.79% market share due to high surgical demand and outpatient care adoption [North America Market Share](#).

On the other hand, Europe benefits from government health initiatives, further strengthening its saline solution market [European Market Growth](#).

Regulatory & Compliance

The regulatory landscape for 0.9% Sodium Chloride Injection, USP, a critical medical solution, highlights its importance and recent market developments. This solution, commonly referred to as saline, is tightly regulated due to its widespread use in replenishing fluids and maintaining electrolyte balance within body compartments [source](#). The product is held to stringent quality and sterility standards to ensure safety, with each batch undergoing rigorous testing [source](#).

Regulatory bodies such as the FDA have played a critical role in addressing supply chain disruptions. Notably, the FDA announced the resolution of a nationwide IV saline shortage, which was previously exacerbated by natural disasters and infrastructure challenges. Increased production and facility restorations by manufacturers were pivotal in overcoming this issue [source](#). This highlights the collaborative efforts between regulatory agencies and producers to ensure supply security.

In the U.S., major manufacturers like Baxter and B. Braun Melsungen AG dominate the market, leveraging partnerships and collaborations for growth [source](#). This competitive dynamic underscores the industry's commitment to both expanding capacity and meeting rigorous regulatory standards, ensuring consistent availability of this vital medical product.

Sourcing & Pricing Channels

The sourcing and pricing landscape for 0.9% saline reflects significant variability depending on the supplier, packaging, and context. For smaller packaging, such as 5 mL unit-dose vials, prices can be as low as \$22.95 for a box of 100 units through distributors like [eBay's alhealth1 store](#), equating to roughly \$0.23 per vial. Meanwhile, larger 1000 mL IV bags carry higher price tags; list prices from distributors range between \$8.19 and \$11.99 per bag, as seen with products supplied by Baxter Healthcare on [BoundTree Medical](#).

Bulk purchasing and institutional contexts introduce even greater variability. While hospitals may procure 1000 mL bags of saline at production costs estimated at \$1.07 per unit—a figure rising from \$0.46 in 2010—charges to patients vary widely, sometimes exceeding \$700 in medical billing due to bundling with ancillary services or hidden fees ([Advisory.com](#)). Suppliers such as [Mountainside Medical](#) also offer tiered pricing for high-volume customers, emphasizing the importance of bulk procurement strategies for cost efficiency.

Overall, as the demand for 0.9% saline spans healthcare facilities, home settings, and laboratories, procurement strategies must weigh price points against logistical considerations like volume, storage, and supplier reliability. The stark discrepancies underlined by both production costs and final consumer billing indicate a dynamic, often opaque market influenced by healthcare system complexities and supply chain factors.

Risks & Alternatives

The use of 0.9% saline, also known as normal saline, carries several risks, particularly for vulnerable patient populations. Among the most significant concerns is [fluid overload](#), which can lead to hypervolemia and exacerbate conditions like congestive heart failure. This issue arises due to the solution's isotonic nature, which traps water in extracellular spaces. Further risks include the potential for [electrolyte imbalances](#) such as hyponatremia and hyperchloremic metabolic acidosis, particularly for patients receiving infusions exceeding 1 liter per day.

Studies also highlight that prolonged use of normal saline increases the likelihood of [metabolic acidosis and acute kidney injuries](#), presumably due to its higher sodium and chloride levels relative to blood. Consequently, careful patient monitoring is essential, especially for individuals with renal insufficiency or those predisposed to edema and sodium retention.

As alternatives to traditional normal saline, research suggests that balanced IV fluids with lower chloride concentrations might reduce risks such as kidney injury and acidosis. Nevertheless, the adoption of these safer alternatives remains limited, potentially due to [cost considerations](#), variations in clinical practice, or logistical constraints. Effective procurement strategies should weigh these risks against the widespread utility and affordability of normal saline in clinical and non-clinical applications.

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